

ChronoArb

MVP Software Requirements Specification

Dealer Acquisition Intelligence for the Secondary Luxury Watch Market

Document Attribute	Value
Document version	1.0
Status	Implementation baseline / MVP
Date	2 August 2026
Target release	Closed dealer beta followed by paid MVP
Primary clients	Next.js web application; Flutter iOS and Android applications
Backend	Python FastAPI modular monolith plus asynchronous ingestion workers
Deployment baseline	Amazon Web Services, managed relational database, queues, object storage and push services

Core product promise — ChronoArb identifies listings that are likely to produce executable, risk-adjusted dealer profit after acquisition costs, condition adjustments, resale expenses and expected time-to-sale. It is not merely an asking-price comparison tool.

CONFIDENTIAL — PRODUCT, ENGINEERING AND OPERATIONS

Document Control

Version	Date	Owner	Summary
0.1	2 Aug 2026	Product / Engineering	Initial implementation-grade specification
1.0	2 Aug 2026	Product / Engineering	MVP baseline covering web, mobile, backend, data, security, DevOps and acceptance

Approvals Required Before Build

Role	Approval Scope
Product owner	Scope, pricing, watch-reference universe, launch metrics
Engineering lead	Architecture, stack, estimates and operational ownership
Data lead	Matching, normalization, valuation and quality methodology
Security owner	Threat model, IAM, secrets, logs and incident response
Legal counsel	Source terms, permitted collection methods, privacy, consumer disclosures
Design lead	Web/mobile information architecture and accessibility

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1. Introduction and Product Scope

1.1 Purpose

This SRS defines the software, data, infrastructure, security and operational requirements for the ChronoArb MVP. It is intended to be sufficiently specific for product design, engineering estimation, implementation, quality assurance, security review and launch approval without requiring a separate architecture document for the MVP.

1.2 Product Vision

ChronoArb will become the acquisition intelligence layer for professional secondary-market watch dealers. It will continuously ingest permitted inventory data, identify and normalize exact watch references and variants, estimate executable acquisition and resale economics, and alert users when a listing exceeds their risk-adjusted profit requirements.

1.3 Business Objectives

- Enable a dealer to find actionable inventory faster than manual marketplace monitoring.
- Produce enough measurable customer value that a professional dealer can justify a \$299 monthly founding plan.

- Build a proprietary outcome dataset connecting public listings to dealer decisions, acquisitions, resale results and realized profit.
- Validate the SaaS model before taking custody of watches, facilitating payments or assuming authentication and inventory risk.
- Create a technical foundation that can later support additional sources, references, enterprise accounts and verified transactions.

1.4 MVP Product Definition

The MVP supports approximately 25 high-liquidity modern watch references and at least three dependable, legally reviewed data integrations. It provides a responsive web dashboard, native Flutter applications for iOS and Android, real-time in-app and push alerts, Telegram alerts, opportunity scoring, user-defined economics, feedback capture, team accounts and subscription billing.

1.5 Explicit Non-Goals

- No automated purchase or browser-driven checkout.
- No escrow, marketplace payments, custody, financing or proprietary inventory.
- No guarantee that a listing is authentic, available or profitable.
- No automated visual authentication of watches.
- No broad vintage-watch coverage during the MVP.
- No unsupported circumvention of access controls, CAPTCHAs, authentication walls or contractual restrictions.
- No public consumer marketplace and no seller listings hosted by ChronoArb.
- No native desktop application; the responsive web app covers desktop workflows.

Product interpretation — “Market value” in this SRS means a documented estimate of expected executable exit value for a specified resale channel and time horizon. It does not mean the median of active asking prices.

2. Stakeholders, Personas and User Journeys

2.1 Stakeholders

Stakeholder	Primary Need	Success Signal
Professional dealer / flipper	Immediate, trustworthy acquisition opportunities	Purchases attributable to alerts and realized profit
Collector	Avoid overpaying and locate target references	Useful target-price alerts and price transparency
Dealer principal / manager	Team productivity and capital allocation	Opportunity pipeline, audit trail and outcome reporting
ChronoArb operations	Healthy source jobs and low false positives	Source SLA, parsing accuracy and rapid remediation
Product and data teams	Labeled decisions and transaction outcomes	Feedback coverage and improving model calibration
Legal and security	Permitted data use and controlled customer data	Reviewed integrations, audit logs and no critical incidents

2.2 Primary Personas

Persona	Profile	Critical Workflow
Dealer	Trades modern Rolex, AP, Patek and adjacent references; time-sensitive; margin-driven	Configure cost model, receive alert, inspect evidence, contact seller, record decision and outcome
Dealer Manager	Runs a team and allocates acquisition capital	Set organization defaults, invite users, review team activity and realized value

Persona	Profile	Critical Workflow
Collector	Tracks a small number of “grail” references	Set target price, receive alert, compare history and open source listing
Operations Analyst	Maintains sources, mappings and data health	Review job failures, unmatched listings, duplicates, source changes and alert quality
System Administrator	Controls plans, entitlements, feature flags and audit access	Administer tenant and platform settings without direct database edits

2.3 Critical User Journeys

1. Onboarding: user creates an account, joins or creates an organization, selects references, configures acquisition costs and connects Telegram and mobile push.
2. Opportunity discovery: a new listing is ingested, normalized and valued; ChronoArb creates an opportunity and matches it against user alert rules.
3. Alert response: user receives push or Telegram notification, opens the opportunity, reviews evidence and economics, and opens the original listing.
4. Decision feedback: user marks purchased, contacted, dismissed, duplicate, unavailable or suspicious and optionally supplies a reason.
5. Realized outcome: a purchasing user records actual acquisition cost, resale proceeds and days-to-sell; the platform calculates realized profit.
6. Operations remediation: a source parser fails or matching confidence falls; operations can disable the source, replay a job or correct a mapping.

3. MVP Boundary and Release Definition

3.1 Included Capability

Capability	MVP Commitment
Watch universe	25 canonical modern references; configuration supports expansion without schema changes
Data integrations	Minimum three approved sources using official APIs, partner feeds or legally reviewed public extraction
Collection frequency	Configurable per source/reference; high-priority inventory polled or received frequently enough to target p95 detection under 10 minutes
Normalization	Reference, variant, currency, geography, condition, box/papers, seller and listing availability
Valuation	Source-aware acquisition estimate, resale channel estimate, cost model, net profit, ROI, liquidity and confidence
Alerts	In-app, Flutter push and Telegram; email optional and disabled by default
Clients	Next.js responsive web app; Flutter iOS/Android app; internal operations UI in web app
Commercial	Stripe web checkout, billing portal, trials, subscription state and entitlements
Operations	Source health, job replay, unmatched queue, audit logs, feature flags and support tooling

3.2 Launch Stages

Stage	Users	Entry Gate	Exit Gate
Internal alpha	Employees and test accounts	Core ingest-to-alert flow works	No blocker defects; traceable calculations
Design-partner beta	10–20 invited dealers	Three sources; 15 references;	Reference accuracy $\geq 97\%$; actionable

Stage	Users	Entry Gate	Exit Gate
		manual operations coverage	feedback collected
Paid MVP	Founding dealers	25 references; billing; mobile apps; operations runbook	Release acceptance criteria in Section 15 satisfied

4. Functional Requirements

Priority definitions: Must = required for paid MVP; Should = included unless a documented launch tradeoff is approved; Could = post-MVP or feature-flagged. All requirements are tenant-isolated unless explicitly marked platform-admin only.

4.1 Identity, Authentication and Sessions

ID	Requirement	Priority	Acceptance / Verification
FR-AUTH-001	The system shall authenticate web and mobile users using OpenID Connect Authorization Code flow; mobile shall use PKCE.	Must	Successful sign-in yields a short-lived access token; invalid, expired or wrong-audience tokens are rejected.
FR-AUTH-002	The system shall support email/password and at least one social or enterprise identity provider through the selected OIDC service.	Should	Configured providers appear in hosted authentication and create one internal user record per stable subject.
FR-AUTH-003	The system shall require verified email before enabling alerts or billing.	Must	Unverified accounts cannot activate an alert rule or start a paid plan.
FR-AUTH-004	The system shall support secure logout from the current device and revocation of refresh sessions.	Must	Logout removes local credentials and prevents token refresh.
FR-AUTH-005	The system shall support optional multi-factor authentication and allow it to be enforced per organization plan.	Should	MFA enrollment, recovery and enforcement are testable in web and mobile.
FR-AUTH-006	The backend shall map OIDC identities to internal user IDs and shall not use email as the immutable identifier.	Must	Changing email does not create a duplicate user.
FR-AUTH-007	Mobile credentials shall be stored only in platform secure storage; web refresh credentials shall be HttpOnly, Secure and SameSite-controlled.	Must	Security review confirms no refresh token in JavaScript-accessible storage or plaintext mobile preferences.

4.2 Organizations, Teams and Roles

ID	Requirement	Priority	Acceptance / Verification
FR-ORG-001	A user shall be able to create one dealer organization during onboarding or accept an invitation to an existing organization.	Must	Organization creation and invitation acceptance produce an active membership.
FR-ORG-002	The system shall support Owner, Admin, Dealer and Viewer roles.	Must	Each protected action is verified against a documented permission matrix.
FR-ORG-003	Owners and Admins shall invite, revoke and change roles for team members.	Must	Invitations expire; revoked users lose tenant access immediately.
FR-ORG-004	Organization settings shall include default base currency, timezone, acquisition costs, resale channel assumptions and notification defaults.	Must	New alert rules inherit defaults and retain an auditable snapshot of assumptions.
FR-ORG-005	All organization-scoped records shall be isolated through backend authorization and database query constraints.	Must	Automated tenant-isolation tests demonstrate that cross-organization IDs return 404 or 403.
FR-ORG-006	The system shall maintain an audit trail for membership, role, billing and organization-setting changes.	Must	Admin can filter immutable audit events by actor, action and time.

4.3 Canonical Watch Catalog

ID	Requirement	Priority	Acceptance / Verification
FR-CAT-001	The system shall maintain canonical brands, collections, references and variants with normalized identifiers.	Must	Each supported listing can reference exactly one canonical watch variant or be marked unmatched.
FR-CAT-002	Catalog records shall support aliases, spacing variants, common abbreviations and source-specific mappings.	Must	Known examples such as "126610 LN" and "126610LN" resolve to one reference.
FR-CAT-003	Catalog records shall store material, dial, bezel, bracelet, movement, production status and liquidity tier where applicable.	Must	Variant fields are exposed through API and visible on detail pages.
FR-CAT-004	Operations users shall create, edit, activate and deactivate references without deployment.	Must	Catalog changes are versioned and reflected in subsequent normalization jobs.
FR-CAT-005	The system shall record matching confidence and matching method: exact, alias, rules, source mapping or reviewed override.	Must	Every normalized listing contains method and confidence or remains unmatched.
FR-CAT-006	Manual mapping overrides shall be auditable and replayable across historical unmatched listings.	Should	Operations can remap and enqueue reprocessing for a selected time range.

4.4 Source Management and Data Acquisition

ID	Requirement	Priority	Acceptance / Verification
FR-SRC-001	Each data source shall be represented by a configurable adapter with access mode, schedule, rate constraints, credentials, legal-review status and health state.	Must	A source cannot run in production until status is Approved.
FR-SRC-002	The preferred access order shall be official API, commercial/affiliate feed, partner feed, permissioned crawl, then legally reviewed public extraction.	Must	Source record documents the selected access mode and approval evidence.
FR-SRC-003	The system shall not implement CAPTCHA solving, credential theft, access-control bypass or collection from authenticated pages without permission.	Must	Code review and source runbook confirm prohibited mechanisms are absent.
FR-SRC-004	The scheduler shall enqueue idempotent collection jobs by source, watch scope and time window.	Must	Duplicate schedule events do not produce duplicate listing records.
FR-SRC-005	Adapters shall support API/JSON, XML/CSV feed and HTML page inputs through a common adapter contract.	Must	At least one automated contract test exists for each supported access type.
FR-SRC-006	Collection jobs shall apply source-specific rate limits, concurrency limits, timeouts, retries and exponential backoff.	Must	Synthetic failure tests respect configured limits and move exhausted jobs to a DLQ.
FR-SRC-007	The system shall retain request metadata, response status, collection timestamp and a content checksum for traceability.	Must	Every snapshot can be tied to a source job and raw evidence object.
FR-SRC-008	Source availability and parser health shall be visible to operations users.	Must	Dashboard shows last success, failure rate, lag, listing count and parser anomaly status.
FR-SRC-009	A source may be paused globally or for a specific adapter version without redeploying the entire platform.	Must	Paused sources stop new jobs while preserving historical data.
FR-SRC-010	Failed and dead-letter jobs shall be replayable by authorized operations users.	Should	Replay produces a new attempt linked to the original job and preserves history.

4.5 Raw Evidence, Parsing and Listing Lifecycle

ID	Requirement	Priority	Acceptance / Verification
FR-LST-001	The system shall create one logical listing per source and stable external listing ID; URL-only sources shall use a deterministic source key.	Must	Repeated observations update the listing rather than creating duplicates.
FR-LST-002	Each fetch shall create an immutable snapshot containing raw evidence location, checksum and parsed fields.	Must	Operations can retrieve the exact evidence used for a valuation, subject to retention.
FR-LST-003	The parser shall extract title, description, price, currency, seller, location, images, condition text, accessories, year, availability and timestamps where present.	Must	Per-source fixtures validate required fields and graceful null handling.
FR-LST-004	The system shall track first seen, last seen, price history and current lifecycle status: Active, Sold, Removed, Unavailable, Unknown.	Must	Status transitions are timestamped and visible in listing history.
FR-LST-005	Listings shall be marked stale when not confirmed within a configurable source-specific interval.	Must	Stale listings are excluded or penalized by valuation and alerts according to policy.
FR-LST-006	Price changes shall trigger re-normalization and opportunity recalculation.	Must	A changed price creates a valuation event and may create/update/close an opportunity.
FR-LST-007	Raw snapshots shall be access-controlled and never rendered as active HTML inside the user application.	Must	Evidence is displayed as escaped text or sanitized fields; raw objects require signed, authorized access.

4.6 Normalization, Condition and Accessories

ID	Requirement	Priority	Acceptance / Verification
FR-NRM-001	All monetary amounts shall be stored in source currency and converted to the organization base currency using a timestamped exchange-rate record.	Must	Opportunity calculation identifies the FX rate and timestamp used.
FR-NRM-002	The system shall normalize geography into country and region and determine whether cross-border shipping, tax or duty assumptions apply.	Must	All-in cost changes when source country differs from organization settings.
FR-NRM-003	The system shall classify condition into Unworn, Mint, Excellent, Good, Fair or Unknown and retain extracted evidence.	Must	Classification includes confidence and source text snippets.
FR-NRM-004	The system shall classify set status as Full Set, Watch and Papers, Watch and Box, Watch Only or Unknown.	Must	Known box/papers phrases in fixtures map to expected classes.
FR-NRM-005	The system shall extract risk attributes including aftermarket parts, polishing, service need, missing links, damage and uncertain authenticity when stated.	Must	Extracted risk flags appear on opportunity detail and influence confidence or reserves.
FR-NRM-006	Users shall see "Unknown" rather than inferred certainty when evidence is insufficient.	Must	Null/unknown values are not silently converted to favorable defaults.
FR-NRM-007	Operations shall be able to correct normalized attributes and mark corrections as training labels.	Should	Override stores actor, reason, previous value and new value.

4.7 Duplicate and Cross-Post Detection

ID	Requirement	Priority	Acceptance / Verification
FR-DUP-001	The system shall detect likely duplicate	Must	Duplicate groups are created with

ID	Requirement	Priority	Acceptance / Verification
	physical watches across sources using seller, text, structured attributes, images and partial serial evidence where available.		confidence and reason codes.
FR-DUP-002	Only one representative listing from a high-confidence duplicate group shall influence market supply and alert generation unless the user requests otherwise.	Must	The same physical watch does not generate multiple default alerts.
FR-DUP-003	A price discrepancy within a duplicate group shall be displayed as a source inconsistency and shall not automatically be treated as arbitrage.	Must	Opportunity detail identifies cross-posted prices and representative selection.
FR-DUP-004	Users and operations shall report false duplicate and missed duplicate decisions.	Should	Feedback updates group status and becomes labeled data.

4.8 Valuation and Cost Model

ID	Requirement	Priority	Acceptance / Verification
FR-VAL-001	The valuation engine shall produce separate active-ask, expected acquisition, expected private-sale, dealer-to-dealer and retail-dealer values where data permits.	Must	API returns value type, range, as-of time, sample size and confidence.
FR-VAL-002	Valuations shall use source-aware and reference-specific rules or models; one universal watch model shall not be required for MVP.	Must	Valuation version identifies the reference model/configuration used.
FR-VAL-003	Completed or confirmed transaction evidence shall receive higher weight than active asking prices.	Must	Documented weighting is unit-tested and exposed in model metadata.
FR-VAL-004	The engine shall adjust for condition, set status, year, geography, seller type, warranty and relevant risk flags.	Must	A controlled fixture demonstrates the expected directional adjustment.
FR-VAL-005	Each organization shall configure buyer fees, tax, duty, shipping, authentication, service reserve, financing cost, selling fees and risk reserve.	Must	Net-profit calculation changes immediately when assumptions change.
FR-VAL-006	The engine shall calculate all-in acquisition cost, expected net resale proceeds, expected net profit, ROI and annualized capital return.	Must	Formula outputs match approved financial test cases within rounding tolerance.
FR-VAL-007	The engine shall estimate expected days-to-sale using liquidity tier and available historical observations.	Should	Opportunity detail presents a range and identifies low-confidence estimates.
FR-VAL-008	The engine shall return valuation ranges and confidence rather than a false-precision single figure.	Must	UI displays low/high range and confidence with explanatory factors.
FR-VAL-009	Every valuation shall be reproducible from versioned inputs, cost assumptions, FX rate and model version.	Must	Support can replay a valuation and obtain the same result for immutable inputs.
FR-VAL-010	The system shall not present estimates as guarantees or authentication conclusions.	Must	Required disclosures appear on opportunity and valuation views.

4.9 Opportunity Generation and Scoring

ID	Requirement	Priority	Acceptance / Verification
FR-OPP-001	An opportunity shall be created when a normalized active listing can be valued and exceeds platform or user-defined minimum quality criteria.	Must	Non-matching or too-low-confidence listings do not create user-visible opportunities.

ID	Requirement	Priority	Acceptance / Verification
FR-OPP-002	Opportunity score shall incorporate expected net profit, ROI, confidence, liquidity, availability, seller risk, duplicate risk and alert freshness.	Must	Score components are visible in explainability output.
FR-OPP-003	Opportunity status shall include New, Active, Contacted, Purchased, Dismissed, Closed, Expired and Invalid.	Must	Status transitions are governed by documented rules and audit events.
FR-OPP-004	Material changes to listing price, availability or valuation shall update the opportunity and may trigger a new alert only when user cooldown rules permit.	Must	Tests prevent repetitive alert storms for minor changes.
FR-OPP-005	The system shall calculate a data-freshness age and penalize or suppress stale opportunities.	Must	Stale age is visible and can invalidate an alert.
FR-OPP-006	Each opportunity shall include a human-readable explanation of why it is attractive and what may invalidate the spread.	Must	Explanation names top positive and negative factors without unsupported claims.

4.10 Alert Rules and Notifications

ID	Requirement	Priority	Acceptance / Verification
FR-ALT-001	Users shall create alert rules by reference or watch list with minimum net profit, ROI, confidence, maximum acquisition price and geography filters.	Must	Matching engine correctly includes and excludes test opportunities.
FR-ALT-002	Users shall select one or more channels: in-app, mobile push and Telegram.	Must	Channel preference is enforced per rule and per device.
FR-ALT-003	Telegram connection shall use a one-time linking flow initiated by the authenticated user.	Must	A Telegram chat cannot be linked without a server-issued expiring token and user initiation.
FR-ALT-004	The platform shall deduplicate notifications by opportunity, rule, channel and material version.	Must	Repeated worker delivery does not create repeated user messages.
FR-ALT-005	Alert messages shall include reference, source, listing price, all-in cost, expected net profit, ROI, confidence, freshness and deep link.	Must	All required fields are present or explicitly marked unavailable.
FR-ALT-006	Users shall configure quiet hours while allowing an override for high-priority opportunities.	Should	Notifications are delayed or sent according to organization timezone and threshold.
FR-ALT-007	Notification delivery status shall be recorded as Queued, Sent, Delivered where supported, Failed or Suppressed.	Must	Operations can search failures and retry eligible notifications.
FR-ALT-008	Device tokens and Telegram identifiers shall be revocable and removed on logout, un-link or repeated permanent delivery failure.	Must	Invalid endpoints stop receiving sends and are marked inactive.
FR-ALT-009	Foreground web and mobile sessions shall receive opportunity updates through WebSocket or equivalent real-time stream.	Should	New opportunities appear without full-page reload under normal network conditions.

4.11 Web Dashboard and User Experience

ID	Requirement	Priority	Acceptance / Verification
FR-WEB-001	The web application shall provide a ranked opportunity feed with filters for reference, source, price, profit, ROI, confidence, freshness, geography and status.	Must	Filters are URL-addressable and survive refresh.
FR-WEB-002	The opportunity detail page shall show listing	Must	All calculations are traceable and original

ID	Requirement	Priority	Acceptance / Verification
	evidence, valuation range, cost waterfall, score explanation, history, duplicate context and actions.		listing opens in a new safe tab.
FR-WEB-003	The watch detail page shall show current value ranges, active comparable listings and 30/90-day historical charts when sufficient data exists.	Must	Charts disclose sample size and gaps; no fabricated 365-day history.
FR-WEB-004	Users shall manage watch lists, alert rules, cost assumptions, notification channels, team settings and billing.	Must	All settings have validation, success confirmation and audit record where relevant.
FR-WEB-005	The interface shall distinguish estimates, observed values and user-entered outcomes visually and semantically.	Must	Accessibility review confirms labels do not depend only on color.
FR-WEB-006	The web app shall include responsive layouts for desktop, tablet and mobile web.	Must	Core workflows pass viewport tests at 360, 768, 1024 and 1440 px.
FR-WEB-007	Operations routes shall be hidden from customer navigation and protected by platform roles.	Must	Direct URL access by customers is denied server-side.

4.12 Feedback and Trade Outcomes

ID	Requirement	Priority	Acceptance / Verification
FR-FBK-001	Users shall provide one-tap opportunity decisions: Purchased, Contacted, Already Sold, Wrong Reference, Condition Issue, No Margin, Suspicious, Duplicate or Not Interested.	Must	Decision is recorded with user, timestamp and optional comment.
FR-FBK-002	Purchased opportunities shall support actual purchase price, fees, date and acquisition channel.	Must	Actual acquisition economics replace estimates only in outcome reporting, not historical prediction records.
FR-FBK-003	Users shall record resale price, selling costs, date and channel to calculate realized profit and days-to-sale.	Must	Realized results are separately labeled from predicted results.
FR-FBK-004	Organizations shall control whether individual trade outcomes are visible to all team members.	Should	Visibility setting is enforced in API and UI.
FR-FBK-005	The platform shall aggregate anonymized outcome metrics only under the privacy and contractual policy approved for the product.	Must	Consent/terms and data-use configuration are documented before aggregation.
FR-FBK-006	Feedback shall be exportable for internal data-quality review with sensitive fields appropriately restricted.	Should	Authorized analyst export contains lineage and excludes secrets or raw identity data not needed.

4.13 Subscription Billing and Entitlements

ID	Requirement	Priority	Acceptance / Verification
FR-BIL-001	Web billing shall use Stripe Checkout or equivalent hosted flow for trial and subscription purchase.	Must	No card data is handled by ChronoArb servers; successful checkout updates entitlements through verified webhook.
FR-BIL-002	The system shall support a founding Professional plan and configuration-driven future tiers.	Must	Plan limits are stored as entitlements, not hard-coded in clients.
FR-BIL-003	Stripe webhook processing shall be signature-verified, idempotent and replayable.	Must	Duplicate and out-of-order events converge to the correct subscription state.
FR-BIL-004	Users shall manage payment method, invoices and cancellation through a hosted billing portal.	Must	Portal access requires authenticated ownership of the billing customer.

ID	Requirement	Priority	Acceptance / Verification
FR-BIL-005	The platform shall enforce grace, past-due, canceled and expired states without deleting customer data immediately.	Must	Entitlements follow an approved state-transition table and user sees remediation guidance.
FR-BIL-006	The mobile MVP shall support sign-in to an existing entitlement. Any in-app purchase or external purchase call-to-action shall be implemented only after current Apple and Google policy review.	Must	App-store submission checklist records the approved billing approach for each storefront and region.

4.14 Internal Operations and Administration

ID	Requirement	Priority	Acceptance / Verification
FR-ADM-001	Platform administrators shall manage sources, adapter versions, watch references, feature flags, plans and customer support state.	Must	All destructive or entitlement-changing actions are audited.
FR-ADM-002	Operations shall review unmatched listings, low-confidence matches, parser anomalies and duplicate conflicts.	Must	Queues support filtering, bulk triage and single-record correction.
FR-ADM-003	Operations shall inspect source job logs, raw evidence metadata, parsed output and processing events from one trace view.	Must	A trace ID connects schedule, fetch, parse, normalize, value and notify events.
FR-ADM-004	Authorized support staff shall impersonate a customer only through a time-limited, visible, audited support session or use read-only diagnostic views.	Should	Impersonation requires elevated authorization and produces prominent banners and audit events.
FR-ADM-005	Feature flags shall support environment, tenant and percentage rollout scopes.	Should	A feature can be disabled without deployment and changes are audited.
FR-ADM-006	The system shall provide a kill switch for alerts and each data source.	Must	Authorized staff can stop outbound alerts or a problematic source within minutes.

4.15 Flutter Mobile Application

ID	Requirement	Priority	Acceptance / Verification
FR-MOB-001	The Flutter app shall support iOS and Android from one codebase while retaining platform-specific secure storage, notification and deep-link integrations.	Must	Release builds pass supported-device test matrix on both platforms.
FR-MOB-002	Mobile shall provide login, onboarding completion, opportunity feed, detail, watch lists, alert rules, feedback, settings and notification inbox.	Must	All listed workflows can be completed without using mobile web, except subscription purchase where policy requires.
FR-MOB-003	Push notifications shall deep-link to the exact opportunity and handle foreground, background and terminated states.	Must	Automated/manual tests cover all app lifecycle states.
FR-MOB-004	The app shall cache the most recent feed and opportunity details for read-only offline access.	Should	Previously viewed data loads without network and clearly shows stale timestamp.
FR-MOB-005	User-entered feedback created offline shall be queued locally and synchronized idempotently when connectivity returns.	Should	Network interruption does not duplicate decisions or outcomes.
FR-MOB-006	The app shall prevent sensitive values from appearing in application-switcher previews where practical and shall support biometric re-entry as an optional local control.	Should	Platform security review verifies implementation behavior.
FR-MOB-007	The app shall expose notification permissions contextually and provide recovery instructions when permissions are denied.	Must	User can reach system settings and app shows actual permission state.

ID	Requirement	Priority	Acceptance / Verification
FR-MOB-008	Crash and performance telemetry shall exclude access tokens, raw listing descriptions containing personal data and user-entered financial details.	Must	Telemetry scrub rules are tested before release.

5. Data Requirements and Domain Model

5.1 Core Data Principles

- Immutable raw evidence and immutable prediction records; corrections create new versions rather than rewriting historical calculations.
- Every customer-visible estimate must carry source time, valuation version, confidence and applicable user cost-assumption version.
- Source data, normalized data, predictions, user decisions and realized outcomes are distinct layers.
- Tenant ownership is explicit on all customer-created records; shared market data is separated from private organization data.
- Prices use decimal fixed-point types; floating point is prohibited for financial calculations.
- Timestamps are stored in UTC and rendered in the user or organization timezone.

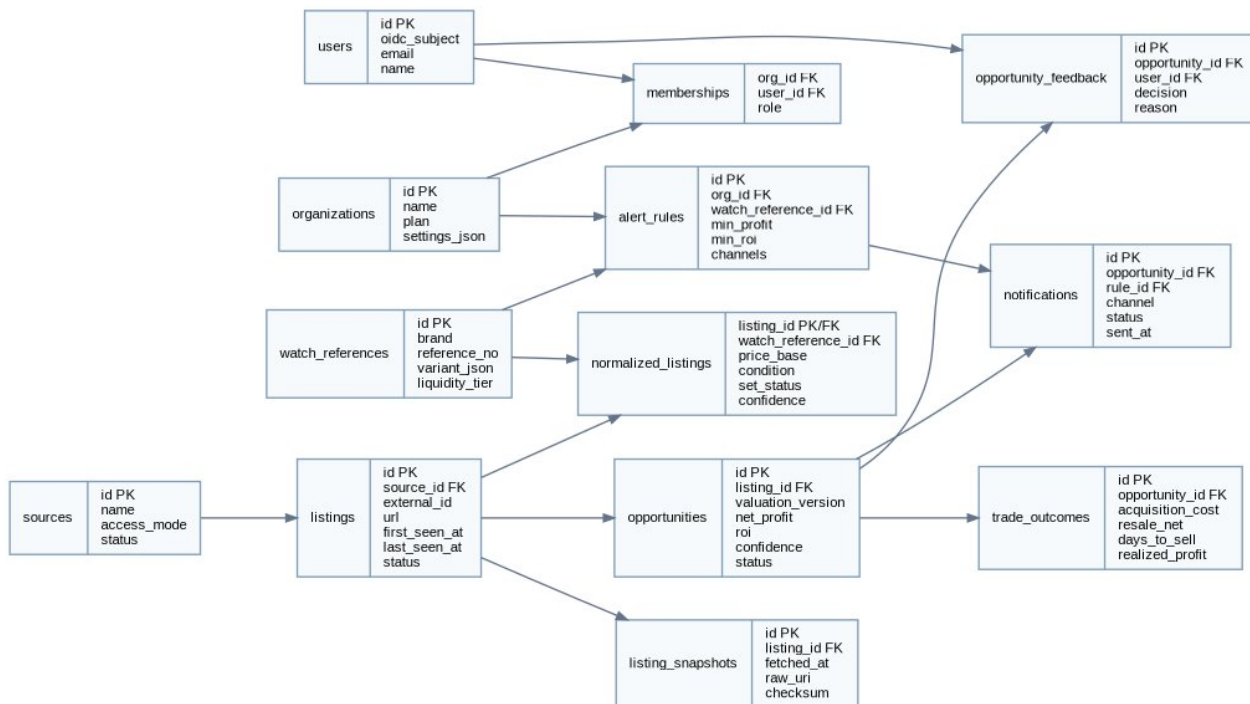


Figure 1. Logical MVP domain model. Physical schema may add partitioning, history and junction tables.

5.2 Principal Entities

Entity	Purpose	Key Notes
organizations	Tenant and dealer configuration	Owns memberships, billing, assumptions and alert rules
users / memberships	Identity and tenant roles	OIDC subject is immutable; membership defines access
watch_references	Canonical watch and variant catalog	Includes aliases, liquidity and activation state
sources / source_jobs	Integration metadata and collection execution	Stores approval state, adapter version, trace and health
listings	Logical external listing lifecycle	Unique by source + external identifier

Entity	Purpose	Key Notes
listing_snapshots	Immutable observations and raw evidence pointer	Content-addressed evidence in object storage
normalized_listings	Latest/ versioned structured interpretation	Reference, condition, set, geography and confidence
duplicate_groups	Cross-posted physical-watch hypotheses	Representative listing plus confidence and evidence
valuations	Versioned expected values and factors	Reference/channel/time-horizon specific
opportunities	User-relevant actionable listing economics	Status and score tied to valuation version
alert_rules / notifications	User criteria and delivery attempts	Idempotency keys and delivery state
feedback / trade_outcomes	Decisions and realized economics	Private by default; separately permissioned
audit_events	Security and material business changes	Append-only, searchable, retention-controlled

5.3 Retention Baseline

Data Class	MVP Retention	Notes
Raw public-source evidence	90 days hot; archive up to 12 months if permitted	Shorter retention overrides apply when source terms require
Parsed and normalized market observations	Minimum 24 months	Needed for trends and model evaluation
Valuations and opportunities	Minimum 24 months	Immutable prediction history
Customer configuration and audit	Account lifetime plus legally approved period	Deletion/anonymization workflow applies
Notification delivery logs	180 days	Payload minimization; delivery troubleshooting
Security logs	Minimum 12 months	Restricted access and centralized retention
Billing records	According to accounting, tax and Stripe obligations	Do not duplicate unnecessary payment data

5.4 Data Quality Rules

Rule	Threshold / Behavior
Reference match	User-visible opportunity requires confidence $\geq$ configured threshold; launch target 0.90 unless source mapping is exact
Currency	Unknown or implausible currency suppresses opportunity
Price	Zero, negative or extreme outlier price is quarantined
Availability	Stale/unconfirmed listing is penalized or suppressed
Duplicate	High-confidence duplicate group yields one default representative
Valuation sample	Low sample size lowers confidence and displays explicit warning
Lineage	No opportunity without listing snapshot, normalizer version and valuation version

6. External Interfaces and API Contract

6.1 Public Application API

The backend exposes versioned JSON REST endpoints under /api/v1 and a WebSocket endpoint for authenticated foreground updates. OpenAPI is generated from the backend models, reviewed in CI and used to generate typed client bindings where practical.

Method / Path	Purpose	Auth
GET /api/v1/me	Current user, memberships and entitlements	User
GET /api/v1/opportunities	Paginated ranked feed and filters	Member
GET /api/v1/opportunities/{id}	Full opportunity, economics, evidence and history	Member
POST /api/v1/opportunities/{id}/feedback	Decision and reason	Dealer
POST /api/v1/opportunities/{id}/outcome	Actual acquisition/resale outcome	Dealer
GET /api/v1/watches	Supported catalog and watch-list state	Member
GET /api/v1/watches/{id}/market	Value ranges, comparables and history	Member
GET/POST/PATCH/DELETE /api/v1/alert-rules	Manage rules	Dealer/Admin
GET/PATCH /api/v1/organization	Tenant settings	Member/Admin
POST /api/v1/organization/invitations	Invite team member	Admin
POST /api/v1/integrations/telegram/link	Create expiring linking token	Member
POST /api/v1/devices	Register push token and device metadata	Member
DELETE /api/v1/devices/{id}	Revoke device endpoint	Member
POST /api/v1/billing/checkout	Create hosted checkout session	Owner
POST /api/v1/billing/portal	Create hosted billing portal session	Owner
POST /api/v1/webhooks/stripe	Verified billing webhook	Stripe signature
WS /api/v1/stream	Opportunity and notification updates	Member

6.2 API Conventions

- JSON uses snake_case; timestamps use RFC 3339 UTC; money uses decimal string plus ISO 4217 currency.
- Pagination uses opaque cursor tokens rather than page numbers for changing feeds.
- Write endpoints accept Idempotency-Key where user retries can occur.
- Errors use a stable envelope: code, message, field_errors, trace_id and retryable.
- ETag or updated_at preconditions protect settings from accidental overwrite where useful.
- Access tokens are validated for issuer, audience, signature, expiry and tenant membership.
- Rate limits are applied by identity, organization, route and IP risk profile.

Example opportunity summary

```
{
  "id": "opp_01J...",
  "watch": {"brand": "Rolex", "reference": "126610LN"},
  "listing_price": {"amount": "11150.00", "currency": "USD"},
  "all_in_acquisition": {"amount": "11480.00", "currency": "USD"},
  "expected_net_profit": {"amount": "1070.00", "currency": "USD"},
  "roi_percent": "9.32",
  "expected_days_to_sell": {"low": 18, "high": 30},
  "confidence": "0.86",
  "freshness_seconds": 92,
  "risk_flags": ["SERVICE_HISTORY_UNKNOWN"],
  "valuation_version": "val-126610ln-2026-08-02.1"
}
```

6.3 Third-Party Interfaces

Interface	MVP Use	Control
Permitted market APIs/feeds	Inventory collection	Source-specific credentials, rate limits, contract tests and legal approval
eBay Browse API or equivalent official endpoint	Search and listing metadata where approved	License compliance, attribution/usage review and API quotas
Telegram Bot API	Instant text alerts and linking flow	HTTPS API, secret token, chat ownership validation and idempotency
Firebase Cloud Messaging / APNs	Cross-platform mobile push	Device tokens, minimal payload and deep-link routing
Stripe Billing	Web subscriptions, portal and entitlements	Hosted UI, signed webhooks and restricted keys
OIDC identity provider	Authentication, MFA and session lifecycle	Authorization Code + PKCE; JWKS validation
FX data provider	Timestamped exchange rates	Daily baseline plus intraday refresh where needed; provider fallback
Sentry / analytics platform	Error and product telemetry	PII scrubbing, region/config review and opt-out where required

7. System Architecture and Technology Stack

7.1 Architecture Style

The MVP shall use a modular monolith for customer-facing domain logic and APIs, deployed independently from asynchronous ingestion and processing workers. All components share a versioned domain package and database schema but have separate processes, queues, scaling policies and failure domains. Microservices are not required until independent ownership or scaling evidence justifies them.

Why this architecture — Scraping and parsing are bursty and failure-prone, while user APIs require predictable latency. Separate workers and queues isolate that workload without introducing the deployment and data-consistency overhead of many microservices.

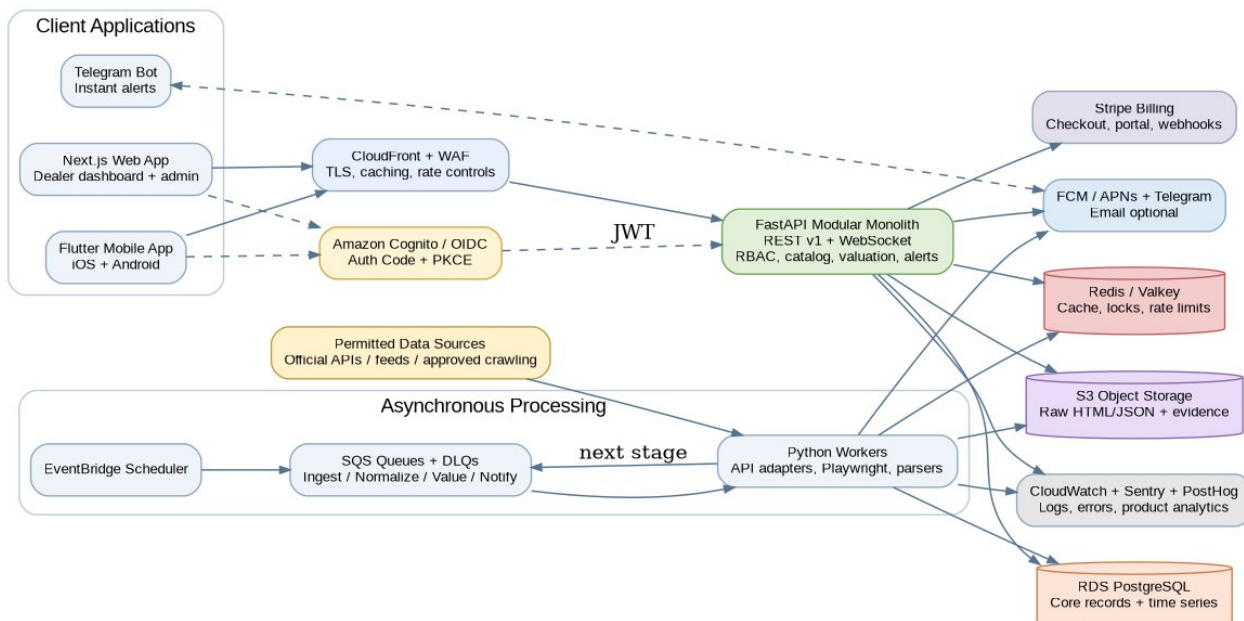


Figure 2. Recommended production architecture for the MVP.

7.2 Recommended Technology Baseline

Layer	Technology	Implementation Baseline / Rationale
Web	Next.js 16.2.x, React 19, TypeScript strict	App Router, server rendering where useful, route handlers only for web-specific concerns; business API remains FastAPI
Web UI	Tailwind CSS, Radix/shadcn-style accessible primitives, TanStack Query, Zod	Fast responsive UI, accessible components, typed server state and validation
Mobile	Flutter 3.44.x and Dart stable	Single iOS/Android codebase with native-quality push, secure storage and deep links
Mobile state	Riverpod, go_router, Dio, freezed/json_serializable	Testable dependency injection, declarative routing and typed models
Backend API	Python 3.13.x, FastAPI 0.139.x, Pydantic 2, SQLAlchemy 2, Alembic	Strong schema generation, async API and mature relational tooling; upgrade to Python 3.14 after dependency CI approval
Collection	HTTPX, Scrapy components, BeautifulSoup/lxml, Playwright only where permitted and necessary	API/feed first; browser rendering isolated in workers
Database	Amazon RDS PostgreSQL 18.4 or approved current RDS minor	Relational integrity, JSONB, window functions, time-based partitioning and mature operations
Cache / locks	ElastiCache Redis OSS 7.1 or Valkey equivalent supported by AWS	Short-lived cache, distributed locks, rate counters and WebSocket fanout; not system of record
Queues	Amazon SQS Standard + DLQs	Durable decoupling, retries, visibility timeout and simple operations
Scheduler	Amazon EventBridge Scheduler	Per-source schedules and controlled fanout
Object storage	Amazon S3 with KMS encryption and lifecycle rules	Raw evidence, exports and immutable snapshots
Identity	Amazon Cognito or equivalent standards-compliant OIDC provider	Shared web/mobile auth, MFA and enterprise path
Push	Firebase Cloud Messaging with APNs configuration	One backend integration for Android and iOS routing
Billing	Stripe Billing, Checkout and Customer Portal	Hosted PCI-sensitive UX and webhook-based entitlements
Cloud runtime	ECS Fargate, ALB, CloudFront, WAF, Secrets Manager and KMS	Container deployment without Kubernetes overhead
Observability	OpenTelemetry, CloudWatch, Sentry and PostHog or equivalent	Trace ingest-to-alert, errors, infrastructure metrics and product events
IaC / CI	Terraform, GitHub Actions, Docker BuildKit, Trivy and dependency scanners	Repeatable environments, security gates and reproducible images

Version note: versions above are the August 2026 implementation baseline and shall be pinned to tested patch releases in lock files and container images. Automated dependency updates require CI and staging validation before production promotion.

7.3 Repository Structure

```

chronoarb/
  apps/
    web/           # Next.js application
    mobile/        # Flutter application
    api/           # FastAPI entry point
    worker/        # ingestion/normalization/valuation/notification workers
  packages/
    domain-python/ # domain services, formulas, policies
    source-adapters/ # adapters and fixtures
    api-client-ts/  # generated TypeScript API client
    design-tokens/ # shared visual tokens and icons
  infrastructure/
    terraform/

```



```
docker/  
docs/  
  adr/          # architecture decision records  
  runbooks/  
  api/  
tests/  
  fixtures/  
  performance/  
  security/
```

7.4 Processing Flow

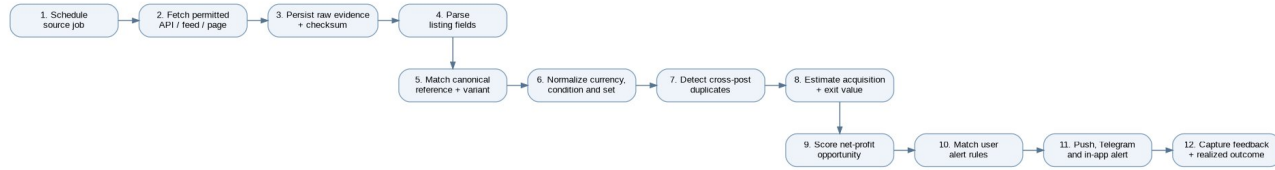


Figure 3. End-to-end listing-to-alert data pipeline.

8. Next.js Web Application Implementation

8.1 Rendering and Application Boundaries

- Use the App Router. Server Components render stable shell, catalog and initial authenticated data where beneficial; highly interactive feeds use Client Components with TanStack Query.
- Use FastAPI as the system-of-record API. Next.js route handlers are limited to web-only edge concerns such as OIDC callbacks, secure cookie exchange and redirects.
- Use TypeScript strict mode, generated OpenAPI types and Zod validation at external boundaries.
- Use server-side authorization for every protected route; client hiding is never treated as access control.
- Use a responsive desktop-first opportunity workflow because professional dealers are expected to perform detailed comparison on large screens.

8.2 Web Route Map

Route	Purpose	Primary Components
/login	OIDC entry and recovery	Hosted auth redirect, support links
/onboarding	Organization, references, assumptions and channels	Stepper with saved progress
/opportunities	Ranked live feed	Filter bar, sortable table/cards, saved views
/opportunities/[id]	Full deal analysis	Cost waterfall, comps, risks, history, actions
/watches	Supported catalog and watch lists	Search, reference cards, tracked status
/watches/[id]	Market detail	Valuation bands, comps and price history
/alerts	Alert rule management	Rule builder, channel preview, test notification
/activity	Decisions and outcomes	Team activity, purchased and dismissed opportunities
/settings/organization	Assumptions and tenant settings	Currency, timezone, fees, roles
/settings/integrations	Telegram and device connections	Link, revoke, delivery test
/settings/billing	Plan and invoices	Checkout/portal launch and entitlement status
/admin/*	Operations console	Sources, jobs, unmatched, duplicates and flags

8.3 Web Component and State Strategy

Concern	Decision
Server state	TanStack Query with cursor pagination, background refresh and optimistic feedback mutations
Local UI state	React state or lightweight store only for cross-component transient UI; avoid duplicating server state
Forms	React Hook Form plus Zod schemas shared with generated API models where practical
Tables	Accessible virtualized table only when row counts justify it; maintain keyboard navigation

Concern	Decision
Charts	Client chart library supporting accessible labels and time-series gaps; data prepared by backend
Real time	Authenticated WebSocket updates invalidate or patch query cache; fallback polling with jitter
Error handling	Route-level error boundaries, trace ID display and retry guidance
Feature flags	Server-resolved entitlements and flags passed into route tree; no security decisions from client flags

8.4 Web Accessibility and Performance

- Target WCAG 2.2 AA for customer and operations surfaces.
- Full keyboard support for feed, filters, dialogs and actions; visible focus and logical tab order.
- Charts provide text summaries and tabular alternatives for critical values.
- Initial authenticated shell target: p75 Largest Contentful Paint under 2.5 seconds on supported broadband; interaction latency target under 200 ms for local controls.
- Use image optimization only for permitted listing thumbnails; do not proxy or retain images beyond source rights.
- Use bundle analysis and route-level code splitting; admin code is not included in ordinary customer routes.

8.5 Web Testing

- Vitest for unit/component logic; Testing Library for user-centric component tests.
- Playwright for end-to-end tests across Chromium, WebKit and Firefox.
- Contract tests generated from OpenAPI; CI fails on unreviewed breaking changes.
- Axe or equivalent automated accessibility checks plus manual keyboard/screen-reader review.
- Visual regression coverage for opportunity, cost waterfall, alert builder and operations error states.

9. Flutter Mobile Application Implementation

9.1 Supported Platforms and Architecture

The mobile application shall target current supported iOS and Android releases according to Flutter and store requirements at build time. The codebase shall use a feature-first clean architecture with presentation, application, domain and infrastructure boundaries. Shared generated API models may be transformed into mobile domain models to keep UI independent of transport details.

```
lib/
  app/
    bootstrap.dart
    router.dart
    theme.dart
  core/
    api/           # Dio client, auth interceptor, generated DTOs
    auth/          # OIDC/PKCE session management
    storage/       # secure storage + encrypted local database
    notifications/ # FCM/APNs, local display, deep links
    telemetry/
  features/
    onboarding/
    opportunities/
    watches/
    alerts/
    activity/
    settings/
  shared/
    widgets/
    models/
    formatters/
  test/
```

integration_test/

9.2 Flutter Package Baseline

Concern	Package / Pattern	Notes
State and DI	flutter_riverpod / Riverpod code generation	Providers scoped by feature and easily overridden in tests
Routing	go_router	Typed/declarative routes, authentication redirects and deep links
Networking	dio	Interceptors, cancellation, timeouts and structured errors
Serialization	freezed + json_serializable	Immutable models and explicit unions for states
Secure tokens	flutter_secure_storage	Keychain/Keystore-backed storage; no tokens in SharedPreferences
Offline cache	drift with SQLCipher-compatible encryption or approved encrypted store	Recent feed, details and pending feedback only
Push	firebase_messaging plus platform APNs setup	Foreground/background/terminated handling
Local notifications	flutter_local_notifications	Consistent display and notification actions where supported
Deep links	app_links or platform routing integration	Opportunity ID routing with auth deferral
Charts	fl_chart or equivalent	Accessible summary provided alongside visuals
Telemetry	sentry_flutter / OpenTelemetry-compatible instrumentation	PII scrubbed before transmission
Testing	flutter_test, mocktail, integration_test, golden tests	Unit, widget, integration and critical visual states

9.3 Mobile Navigation

- Bottom navigation: Opportunities, Watches, Activity and Settings.
- Alert Rules accessible from Opportunities and Settings; notification inbox accessible from app bar.
- Deep link route /opportunities/:id opens after authentication and organization resolution.
- On small devices, cost breakdown and valuation factors use progressive disclosure rather than dense tables.
- External source links open the system browser with confirmation and a safe URL policy.

9.4 Push Notification Design

1. Backend sends a minimal push payload containing notification ID, opportunity ID, title, summary and deep-link route. Sensitive full economics are fetched after authentication.
2. App records notification open and fetches the latest opportunity version rather than trusting stale payload values.
3. Foreground notifications update the feed and may show an in-app banner according to user preference.
4. Background and terminated opens route through a pending-link coordinator after session restoration.
5. Device token rotation is sent to the backend; logout revokes the device registration.
6. Notification categories may support “Open”, “Dismiss” and “Contacted” only after safe idempotent action handling is implemented.

9.5 Offline and Synchronization Rules

Data	Offline Behavior
Opportunity feed	Cache last successful pages and timestamp; read-only when offline
Opportunity detail	Cache recently viewed records and evidence summary; never cache raw source HTML
Feedback	Queue with client-generated idempotency key; sync in order when online
Trade outcomes	Draft locally, submit only after explicit user confirmation and network availability

Data	Offline Behavior
Alert rules/settings	Read cached state; edits require connectivity for MVP
Authentication	No offline access after token/session expiry; biometric unlock does not replace server authentication

9.6 Mobile Release Controls

- Separate dev, staging and production bundle identifiers and push credentials.
- Remote feature flags must not enable unreviewed purchasing or payment behavior.
- App-store privacy labels/data safety forms must match actual SDK collection.
- Automated screenshots and metadata shall avoid promises of guaranteed profit.
- Billing links and purchase flows shall be re-reviewed against current storefront and regional policies before each relevant release.

10. Backend, Ingestion and Valuation Implementation

10.1 FastAPI Domain Modules

Module	Responsibilities
identity	JWT validation, user provisioning, memberships and RBAC
catalog	Brands, references, variants, aliases and watch lists
sources	Source configuration, adapters, schedules, job lifecycle and health
listings	Logical listings, snapshots, status and price history
normalization	Reference match, attributes, geography, FX and confidence
duplicates	Candidate generation, similarity scoring and representative selection
valuation	Comparable selection, adjustments, costs, ranges and model versions
opportunities	Scoring, lifecycle, explanations and user views
alerts	Rule matching, cooldowns, channel routing and delivery status
feedback	Decisions, actual trades and outcome metrics
billing	Stripe customer mapping, webhook state and entitlements
operations	Admin queues, feature flags, audits and support tools

10.2 Source Adapter Contract

```
class SourceAdapter(Protocol):
    source_key: str
    adapter_version: str

    async def discover(self, scope: SourceScope) -> AsyncIterator[SourceItemRef]: ...
    async def fetch(self, item: SourceItemRef) -> RawObservation: ...
    def parse(self, raw: RawObservation) -> ParsedListing: ...
    def stable_external_id(self, parsed: ParsedListing) -> str: ...
    def health_assertions(self, batch: ParsedBatch) -> list[HealthIssue]: ...

# Contract requirements
# - deterministic IDs and parser output for identical evidence
# - fixture tests with saved permitted responses
# - no source-specific logic inside core valuation modules
# - explicit rate/concurrency policy and legal approval reference
```

10.3 Queue Topology

Queue	Producer	Consumer	Failure Policy
source-discovery	EventBridge / admin replay	Discovery workers	Retry, then DLQ with source pause threshold
source-fetch	Discovery workers	Fetch workers	Source-specific retry/backoff and idempotent snapshot
normalize-listing	Fetch/parser workers	Normalization workers	Quarantine invalid or low-confidence records
value-listing	Normalizer / price change	Valuation workers	Retry deterministic transient failures; flag model/config errors
match-alerts	Opportunity service	Alert matcher	Idempotent by opportunity material version
send-notification	Alert matcher	Channel workers	Provider retry, permanent-failure suppression and DLQ
analytics-events	API/workers	Analytics sink	Non-blocking; loss monitored but must not block core workflow

10.4 Matching Strategy

1. Normalize Unicode, punctuation, whitespace and common brand/reference separators.
2. Extract exact reference candidates with brand-specific regex and source-specific structured fields.
3. Resolve exact canonical and alias matches.
4. Apply variant constraints for material, dial, bracelet and generation.
5. Use a lightweight supervised classifier only for unresolved cases after sufficient labeled examples exist.
6. Quarantine conflicting high-value matches for operations review.
7. Retain candidate set, features, method and confidence for explainability.

10.5 Valuation Computation

```

all_in_acquisition =
    listing_price
    + buyer_fee
    + tax_and_duty
    + insured_shipping
    + authentication_cost
    + service_reserve
    + financing_cost
    + acquisition_risk_reserve

expected_net_resale =
    expected_exit_price
    - selling_fee
    - outbound_shipping
    - resale_payment_cost
    - exit_risk_reserve

expected_net_profit = expected_net_resale - all_in_acquisition
roi = expected_net_profit / all_in_acquisition
annualized_return = (1 + roi) ** (365 / expected_days_to_sell) - 1

```

Annualized return shall be hidden or strongly caveated for very short holding periods or low-confidence time-to-sale estimates because simple annualization can exaggerate practical performance.

10.6 Initial Valuation Method

- Use robust reference-specific comparable selection with recency decay and source quality weights.
- Use completed/confirmed outcomes where licensed and available; otherwise estimate source ask-to-exit discount from dealer feedback and observed lifecycle.
- Apply rules-based condition, set, year, seller and geography adjustments; version every adjustment table.
- Use median and median absolute deviation or winsorized ranges to resist extreme asks.

- Return low/high bands and confidence based on sample size, dispersion, match quality and observation freshness.
- Keep model evaluation reference-specific; aggregate metrics alone can hide catastrophic errors on expensive references.

11. Security, Privacy and Compliance

11.1 Security Baseline

The MVP shall target OWASP ASVS 5.0 Level 2 controls appropriate to an authenticated SaaS application, with higher scrutiny for authentication, authorization, cryptography, webhooks, billing, administrative functions and supply chain. The product handles commercially sensitive acquisition behavior and must treat tenant isolation as a critical security boundary.

Control Area	Required Baseline
Identity	OIDC, MFA capability, short access tokens, PKCE mobile flow, session revocation
Authorization	Deny-by-default RBAC, organization scoping in every query/service, admin separation
Transport	TLS 1.2+ with modern configuration; HSTS on production web domains
Stored data	RDS, S3 and backups encrypted with KMS; mobile secure storage; no plaintext secrets
Secrets	AWS Secrets Manager; no secrets in repository, images, mobile bundle or logs
Webhooks	Signature verification, timestamp tolerance, replay/idempotency controls
Input/output	Schema validation, output encoding, SSRF protection, safe external-link policy
Admin	MFA enforced, least privilege, audit logging and time-limited elevated access
Supply chain	Pinned dependencies, SBOM, vulnerability scanning, signed images and patch SLA
Incident response	Severity matrix, on-call ownership, containment runbook and customer notification process

11.2 Scraping and Data-Source Compliance

- Every source requires a recorded legal/business review before production enablement, including terms, robots directives where relevant, rate limits, data rights, caching, display and retention.
- Official APIs and negotiated feeds are preferred. Technical feasibility does not constitute permission.
- The product shall not claim affiliation with or endorsement by any marketplace unless contracted.
- Source URLs, images, descriptions and seller information shall be displayed only to the extent permitted and necessary.
- A source can be disabled rapidly in response to a complaint, contract change or technical issue.
- The operations runbook shall include a takedown/contact process and evidence preservation.

11.3 Privacy

Data	Treatment
User identity	Collect minimum account and organization fields; provide access, correction and deletion workflow
Dealer strategies and outcomes	Private tenant data; restricted staff access; no sale to other dealers
Push/Telegram identifiers	Use only for delivery; revoke on unlink and minimize retention
Location	Organization/source geography only; do not collect continuous user GPS
Telemetry	Pseudonymous IDs where possible; scrub tokens, listing raw text and financial outcome details
Raw source seller data	Minimize, restrict, retain only as permitted and avoid unnecessary person-level enrichment

11.4 Fraud and Abuse Controls

- Rate limit account creation, invitations, source-link clicks, API reads and notification tests.

- Detect shared credentials and anomalous scraping of ChronoArb's own dataset without blocking legitimate dealer-team use.
- Protect high-value aggregate data exports behind enterprise entitlement and audit.
- Do not allow customer-supplied arbitrary URLs to be fetched by backend workers in MVP.
- Use allowlisted source domains and hardened egress policy to reduce SSRF and data exfiltration risk.
- Provide a report-suspicious-listing action without representing ChronoArb as an authentication service.

12. Non-Functional Requirements

12.1 Performance and Timeliness

ID	Requirement	Priority	Acceptance / Verification
NFR-PERF-001	Authenticated read API latency shall be p95 ≤500 ms for ordinary cached/list queries and p95 ≤1.5 s for opportunity detail excluding external links.	Must	Load test under expected paid-MVP concurrency.
NFR-PERF-002	WebSocket foreground updates shall reach connected clients within 5 seconds of committed opportunity publication under normal conditions.	Should	End-to-end staging timing test.
NFR-PERF-003	Notification jobs shall begin provider dispatch within 30 seconds of rule match at p95.	Must	Queue and provider timing dashboard.
NFR-PERF-004	Time-to-detect target shall be <3 minutes for push/API feeds where source capability allows and <10 minutes p95 across MVP high-priority sources.	Must	Measured from source-observed/listed timestamp where available, otherwise first retrievable observation.
NFR-PERF-005	The opportunity feed shall support at least 100,000 active records platform-wide without full-table scans in user paths.	Must	Query plan and load test.
NFR-PERF-006	Mobile cold-start target shall be under 3 seconds p75 on supported mid-tier devices, excluding first authentication redirect.	Should	Release performance test matrix.

12.2 Availability, Reliability and Recovery

ID	Requirement	Priority	Acceptance / Verification
NFR-REL-001	Paid MVP customer-facing API and web availability target shall be 99.5% monthly, excluding announced maintenance.	Must	Cloud monitoring and monthly SLO report.
NFR-REL-002	Queue consumers shall be idempotent and safe under at-least-once delivery.	Must	Chaos/retry tests show no duplicate logical records or alerts.
NFR-REL-003	RDS automated backups and point-in-time recovery shall be enabled; RPO target ≤15 minutes and RTO target ≤4 hours.	Must	Quarterly restore drill and documented result.
NFR-REL-004	S3 raw evidence shall use versioning or object-lock controls where appropriate and lifecycle policies.	Should	Infrastructure policy test.
NFR-REL-005	A single source or parser failure shall not prevent API availability or processing of other sources.	Must	Failure injection test.
NFR-REL-006	Critical alert and source kill switches shall function without code deployment.	Must	Operational game-day verification.

12.3 Security and Privacy

ID	Requirement	Priority	Acceptance / Verification
NFR-SEC-001	No Critical or High exploitable vulnerability may	Must	Security scan and review sign-off.

ID	Requirement	Priority	Acceptance / Verification
	be present at production release; Medium findings require owner and remediation date.		
NFR-SEC-002	All privileged and material customer actions shall create tamper-evident audit events.	Must	Audit coverage test and restricted access review.
NFR-SEC-003	Production access shall use SSO/MFA and least-privilege roles; shared cloud credentials are prohibited.	Must	IAM review.
NFR-SEC-004	Logs and telemetry shall be tested for secret and PII leakage.	Must	Automated redaction tests and manual sample review.
NFR-SEC-005	Tenant-isolation tests shall run in CI and before each production release.	Must	Cross-tenant access suite passes.
NFR-SEC-006	Customer deletion/anonymization workflow shall meet the approved privacy policy while preserving legally required billing/audit records.	Must	Staging deletion test and data map review.

12.4 Usability, Accessibility and Compatibility

ID	Requirement	Priority	Acceptance / Verification
NFR-UX-001	A new dealer shall configure the first useful alert rule within 10 minutes without staff assistance.	Should	Moderated onboarding test with target users.
NFR-UX-002	Web shall target WCAG 2.2 AA; mobile shall support VoiceOver/TalkBack, dynamic text and adequate touch targets.	Must	Automated and manual accessibility review.
NFR-UX-003	Web shall support current and previous major versions of Chrome, Safari, Firefox and Edge at release.	Must	Cross-browser E2E suite.
NFR-UX-004	Mobile shall support the OS versions required by current Flutter/store policy and at least the previous two major consumer OS versions where feasible.	Should	Device/cloud test matrix documented per release.
NFR-UX-005	Financial figures shall use locale-aware formatting while retaining explicit currency codes in ambiguous contexts.	Must	Localization tests.
NFR-UX-006	The UI shall clearly disclose data age, confidence and estimate status.	Must	Design review and content QA.

12.5 Maintainability and Scalability

ID	Requirement	Priority	Acceptance / Verification
NFR-MNT-001	Domain modules shall have explicit interfaces and may not import UI or source-specific adapter code.	Must	Architecture lint/review.
NFR-MNT-002	Source parsers shall use saved fixtures and contract tests so source changes are detected before silent data corruption.	Must	CI fixture suite and anomaly alert.
NFR-MNT-003	Database migrations shall be forward-only in production with tested rollback/roll-forward procedures for application compatibility.	Must	Staging migration rehearsal.
NFR-MNT-004	Infrastructure shall be defined through Terraform and reviewed through pull requests.	Must	Drift detection and CI plan.
NFR-MNT-005	The architecture shall scale workers independently by queue depth and source workload.	Must	Autoscaling test or documented capacity exercise.
NFR-MNT-006	A new well-structured source adapter should be implementable without modifying valuation	Should	Adapter template and engineering review against one new-source spike.

ID	Requirement	Priority	Acceptance / Verification
	or user-facing modules.		

13. DevOps, Observability and Operations

13.1 Environments

Environment	Purpose	Data Policy
Local	Developer workflow with Docker dependencies and saved source fixtures	Synthetic/test data only; no production secrets
Development	Shared integration and feature testing	Synthetic plus explicitly approved sandbox data
Staging	Production-like release candidate and performance/security checks	Sanitized or separately collected staging data
Production	Customer service and approved source jobs	Strict access, backups, monitoring and change control

13.2 CI/CD Pipeline

1. Format, lint and type-check Python, TypeScript and Dart.
2. Run unit, component, parser fixture and domain formula tests.
3. Generate OpenAPI and fail on unapproved breaking changes.
4. Run dependency, secret, container and infrastructure security scans.
5. Build immutable images and signed mobile/web artifacts.
6. Deploy to staging, run database migration rehearsal, E2E, accessibility and smoke tests.
7. Require manual production approval from engineering owner; deploy with rolling or blue/green strategy.
8. Run production smoke checks and automatic rollback for failed health thresholds.

13.3 Observability Model

Signal	Examples	Action
Business	Opportunities/day, alert matches, opens, feedback, purchases, realized profit	Product and model review
Data quality	Match confidence, unmatched rate, duplicates, stale listings, value dispersion	Operations triage
Source health	Fetch success, parse count, schema anomaly, lag, HTTP status, quota	Pause/replay/escalate source
Pipeline	Queue depth, oldest message age, processing latency, DLQ count	Autoscale or incident response
Application	API latency/error rate, WebSocket connections, mobile crash-free sessions	Engineering SLO response
Security	Auth failures, admin actions, suspicious access, webhook failures	Security triage
Cost	Per-source compute/proxy/API cost, storage, notification and cloud spend	Unit-economics review

13.4 Required Runbooks

- Source parser failure or material schema change.
- Alert storm, incorrect valuation or kill-switch activation.
- Queue backlog and worker saturation.
- Cross-tenant access suspicion or credential compromise.
- Stripe webhook outage or entitlement mismatch.

- Push/Telegram provider delivery degradation.
- Database restore and regional service disruption.
- Data-source complaint, takedown or legal hold.

14. Testing Strategy

14.1 Test Pyramid

Level	Coverage
Unit	Financial formulas, scoring, permissions, parsers, matching rules and state transitions
Property-based	Money calculations, idempotency keys, normalization invariants and pagination
Contract	Source adapter fixtures, OpenAPI clients, Stripe/Telegram/FCM interfaces
Integration	PostgreSQL, S3, queues, Redis, OIDC token validation and webhook processing
End-to-end	Onboarding, alert-rule creation, opportunity open, feedback, billing and admin triage
Mobile integration	Push deep links, offline queue, session restore and platform permissions
Performance	Feed/API load, queue throughput, source concurrency and WebSocket fanout
Security	SAST/DAST, dependency scanning, tenant isolation, auth abuse, SSRF and webhook replay
Model/data	Reference-level precision, calibration, outlier cases, stale/duplicate suppression
Operational	Restore drills, kill switches, DLQ replay and incident game days

14.2 Required Financial Test Case

```

Given:
  listing price           = 11,150.00 USD
  buyer fee              =    0.00
  tax/duty                =   100.00
  inbound shipping        =    80.00
  authentication          =    50.00
  service/risk reserve    =   100.00
  expected exit price     = 13,150.00
  selling/payment fees    =   450.00
  outbound shipping       =   150.00

Then:
  all-in acquisition      = 11,480.00
  expected net resale     = 12,550.00
  expected net profit     =   1,070.00
  ROI                    =    9.32% (rounded)

```

14.3 Model Acceptance by Reference

Overall false-positive rate is insufficient as the sole measure because one incorrect high-value Patek or AP classification can be more damaging than several low-value misses. Release evaluation shall report each supported reference separately and include at least reference match precision, condition/set extraction accuracy, duplicate precision, alert precision and valuation error where outcome evidence exists.

15. MVP Release Acceptance Criteria

Area	Paid MVP Gate
Scope	25 active supported references and three production-approved sources

Area	Paid MVP Gate
End-to-end	New source observation can produce a normalized, valued, alerted opportunity with complete lineage
Reference quality	≥98% precision on reviewed supported-reference test set; no known systematic variant confusion
Hard false positives	<15% during beta and trending below 10%; no unresolved critical calculation defect
Duplicate quality	≥95% precision on high-confidence duplicate suppression test set
Timeliness	p95 source-to-alert <10 minutes for high-priority supported sources; provider dispatch p95 <30 seconds after match
Availability	Customer API/web staging soak meets 99.5% target design and recovery controls
Security	No open Critical/High release-blocking finding; tenant-isolation suite passes
Clients	Web plus signed iOS and Android release candidates complete critical journeys
Billing	Trial/paid/cancel/past-due entitlement states verified end-to-end
Operations	Source pause, alert kill switch, DLQ replay and restore procedures tested
Observability	Trace ID links source job through notification; required dashboards and alerts active
Legal/product	Source approvals, terms, privacy policy, disclaimers and app-store billing approach signed off
Commercial	At least 10 design partners used the system; paid conversion hypothesis and support process documented

16. Delivery Plan and Post-MVP Roadmap

16.1 Indicative 16-Week Build Plan

Weeks	Delivery
1–2	Architecture baseline, source/legal inventory, UX flows, catalog schema, cloud foundation and CI
3–5	Identity, organizations, canonical catalog, first source adapter, raw snapshots and operations trace
6–8	Normalization, duplicate baseline, cost model, valuation v1, opportunities and Telegram alerts
9–10	Next.js dealer dashboard, settings, feedback and admin queues
11–12	Flutter app, push/deep links, mobile feed/detail and secure session handling
13	Stripe billing, entitlements, onboarding and audit completion
14	Second/third source integrations, performance and model tuning
15	Closed beta hardening, security, accessibility and store submission
16	Paid MVP release candidate, runbooks, restore/kill-switch drills and launch

The schedule assumes a cross-functional team with approximately two backend/data engineers, one web engineer, one Flutter engineer, one product designer, fractional QA/security/DevOps support and active dealer design partners. Source-contract or access delays can dominate the schedule and must be treated as a separate commercial workstream.

16.2 Deferred V1.1 / V2

- Additional references and regional sources based on revenue per reference.
- Enterprise SSO, advanced team workflows and data exports/API access.
- Improved image similarity and supervised condition extraction.
- Portfolio and inventory tracking for dealers.
- Negotiation support and seller communication templates.

- Verified-opportunity workflow with independent authentication and insured logistics.
- Stripe Connect or other marketplace payment architecture only after a separate legal, risk and custody program.
- Automated purchase or balance-sheet inventory only after strict capital, fraud and operational review.

17. Key Risks and Mitigations

Risk	Impact	Mitigation / Trigger
Source access changes	Inventory coverage or detection speed degrades	Prefer contracts/APIs, adapter isolation, diversify sources, kill switch and fallback feeds
Asking-price bias	False opportunity and customer loss	Executable exit values, source discounts, feedback and realized outcome calibration
Wrong reference/variant	Severe false positive on high-value watch	Reference-specific rules, confidence threshold, manual review and per-reference quality gates
Condition hidden in photos/text	Margin disappears after inspection	Risk reserves, Unknown states, evidence snippets and no authentication claim
Cross-post duplicates	Inflated supply and repeated alerts	Image/text/seller duplicate groups and representative suppression
Stale/phantom listing	User wastes time or misses deal	Availability score, freshness, user feedback and source-specific stale policy
Notification latency	Dealer loses time-sensitive deal	Priority queues, provider monitoring, Telegram + push redundancy and alert SLO
Subscription churn	Users do not capture value quickly	Design-partner onboarding, narrow liquid references, proof-of-value reporting and annual plan only after retention
Data leakage	Dealer strategy or outcome exposed	Tenant isolation, staff least privilege, audit and security testing
Mobile billing rejection	Store launch delay	Existing-entitlement companion approach and current policy/legal review before submission
Cloud/source cost growth	Poor SaaS gross margin	Cost per source/reference dashboard, adaptive schedules and revenue-based coverage decisions
Overbuilding marketplace features	Capital and legal risk distract from SaaS	Formal stage gate; brokerage excluded from MVP

Appendix A. Role Permission Summary

Action	Owner	Admin	Dealer	Viewer	Platform Ops
View opportunities	Yes	Yes	Yes	Yes	Support scope
Create alert rules	Yes	Yes	Yes	No	No
Record trade outcomes	Yes	Yes	Yes	No	No
Edit organization assumptions	Yes	Yes	No	No	No
Invite/change members	Yes	Yes	No	No	No
Manage billing	Yes	Optional	No	No	No
Manage source/catalog	No	No	No	No	Yes
Replay jobs / kill switch	No	No	No	No	Restricted
View private outcomes	Configurable	Configurable	Own/configurable	No	Restricted support

Appendix B. Opportunity Explanation Factors

Positive Factors	Negative / Risk Factors
Price materially below expected acquisition benchmark	Condition or set status unknown

Positive Factors	Negative / Risk Factors
Full set and recent manufacturer warranty	Aftermarket parts or service need stated
High-liquidity reference and strong comparable sample	Low sample size or wide market dispersion
Verified active checkout / fresh listing	Stale, unavailable or long-unconfirmed listing
Established seller and clear returns	New/unknown seller or restrictive return policy
Domestic transaction with low logistics cost	Cross-border tax, duty and shipping uncertainty
Distinct listing, not cross-post duplicate	Likely duplicate with inconsistent price
Expected quick exit channel	Slow expected time-to-sale or weak dealer demand

Appendix C. Event and Idempotency Examples

Event: opportunity.published.v1

```
{
  "event_id": "evt_01J...",
  "occurred_at": "2026-08-02T16:42:19Z",
  "trace_id": "trc_01J...",
  "opportunity_id": "opp_01J...",
  "material_version": 3,
  "listing_id": "lst_01J...",
  "valuation_id": "val_01J..."
}
```

Notification idempotency key:

sha256(organization_id + user_id + rule_id + opportunity_id + material_version + channel)

Appendix D. Definitions

Term	Definition
Active ask	Current seller asking price, not necessarily executable transaction value
All-in acquisition	Purchase price plus all expected costs and reserves required to own a saleable watch
Canonical reference	ChronoArb's normalized identity for an exact watch reference/variant
Confidence	Calibrated indication of evidence/model reliability, not probability of guaranteed profit
Executable exit value	Expected price obtainable through a stated channel and time horizon
Hard false positive	Alert invalidated by wrong reference, duplicate, incorrect price/currency, unavailable listing or omitted material condition
Material version	Opportunity revision that changes economics or decision-relevant risk enough to permit re-alerting
Naked / watch only	Watch offered without box or papers; exact market terminology and adjustment vary
Risk-adjusted profit	Expected net profit after explicit costs and uncertainty reserves
Time-to-detect	Elapsed time from source listing availability/observable timestamp to ChronoArb alert receipt

Appendix E. Source and Technology References

Reference	Relevance	URL
[R1] Next.js documentation	Next.js 16.2.12 was the latest documented release at specification date.	https://nextjs.org/docs
[R2] Flutter release notes	Flutter 3.44 is listed as a stable release.	https://docs.flutter.dev/release/release-notes
[R3] Python documentation	Python 3.14.6 was current; the SRS selects 3.13.x initially for conservative dependency compatibility.	https://docs.python.org/
[R4] FastAPI documentation	FastAPI documentation and version guidance recommend pinning a known working release.	https://fastapi.tiangolo.com/
[R5] PostgreSQL documentation	PostgreSQL 18 is current; AWS RDS supports current 18.x minors.	https://www.postgresql.org/docs/current/
[R6] AWS RDS PostgreSQL release calendar	Managed service version support and dates.	https://docs.aws.amazon.com/AmazonRDS/latest/PostgreSQLReleaseNotes/postgresql-release-calendar.html
[R7] AWS ElastiCache engine versions	Redis OSS/Valkey managed compatibility.	https://docs.aws.amazon.com/AmazonElastiCache/latest/dg/engine-versions.html
[R8] Playwright Python	Browser automation and multi-browser testing capabilities.	https://playwright.dev/python/docs/intro
[R9] Firebase Cloud Messaging for Flutter	Official Flutter push integration guide.	https://firebase.google.com/docs/cloud-messaging/flutter/get-started
[R10] Telegram Bot API	HTTP-based bot interface for notification delivery.	https://core.telegram.org/bots/api
[R11] Stripe Billing subscriptions	Hosted subscription and webhook lifecycle documentation.	https://docs.stripe.com/billing/subscriptions/overview
[R12] Stripe Connect	Required future platform/marketplace architecture if ChronoArb later moves money between parties.	https://docs.stripe.com/connect
[R13] OWASP ASVS	Application Security Verification Standard version 5.0.	https://owasp.org/www-project-application-security-verification-standard/
[R14] Apple App Review Guidelines	Current in-app purchase, multiplatform and enterprise/companion app rules must be reviewed for mobile billing.	https://developer.apple.com/app-store/review/guidelines/
[R15] Google Play policy and subscriptions	Current regional billing and subscription rules must be reviewed before release.	https://support.google.com/googleplay/android-developer/answer/12154973
[R16] eBay Browse API	Official marketplace search interface and license-controlled integration path.	https://developer.ebay.com/api-docs/buy/browse/overview.html

Final implementation rule — When a requirement conflicts with a data-source agreement, app-store rule, payment-provider rule or applicable law, the legal/compliance constraint takes precedence. The affected feature must be disabled or redesigned rather than bypassed.